MLOPS

Model Serving Visual Guide

From trained model to production:
batch vs real-time, vLLM, batching, autoscaling.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Batch vs Real-time

Two serving patterns, different SLOs.

Batch

score millions overnight. Throughput matters; latency doesn't.

Real-time

score one request in milliseconds. Latency SLOs dominate.

Default to batch when the decision isn't needed instantly — it is 10-100x cheaper per prediction.

STEP 2/6

Model Servers

Purpose-built inference runtimes.

NVIDIA Triton

Multi-framework server: TensorRT, ONNX, PyTorch — dynamic batching built in.

vLLM

The LLM server: PagedAttention + continuous batching for high-throughput generation.

ONNX / TensorRT

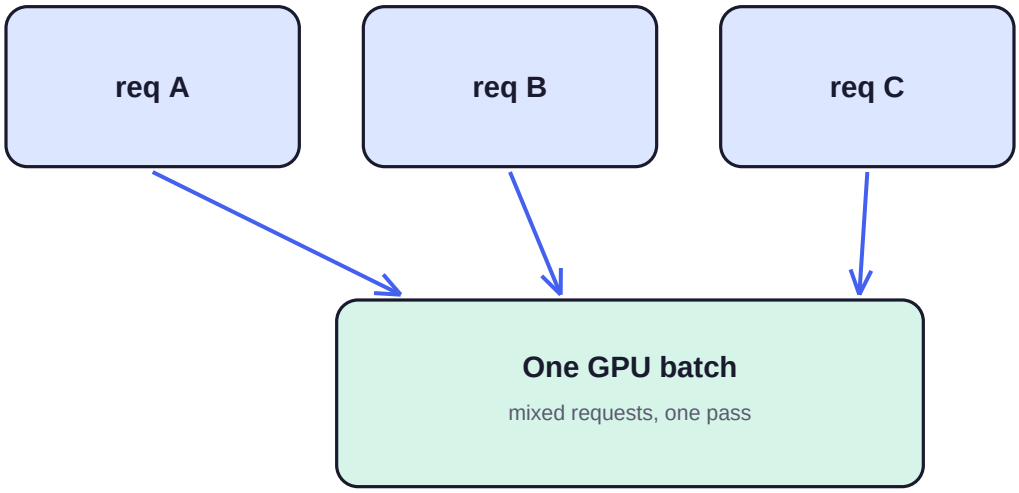
Optimized runtimes: graph fusion and kernel autotuning cut latency sharply.

Don't serve from a Flask app wrapping torch.load — you lose batching, caching, and GPU utilization.

STEP 3/6

Continuous Batching

The trick behind fast LLM APIs.



Requests join and leave the batch token-by-token — no waiting for the slowest request to finish.

Static batching wastes GPU on padding and stragglers. Continuous batching keeps utilization near 100%.

STEP 4/6

The KV Cache

Why the first token is slow and the rest are fast.

Prefill

process the whole prompt: compute-bound, caches keys and values.

Decode

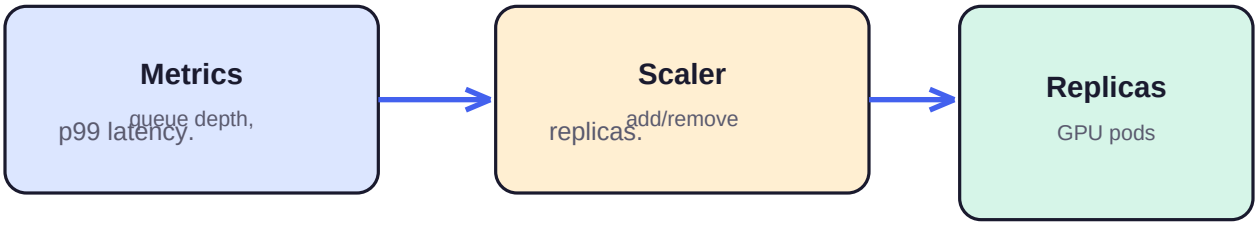
one token at a time, reading the cache — memory-bandwidth bound.

Long contexts explode KV-cache memory. PagedAttention (vLLM) manages it like OS virtual memory.

STEP 5/6

Autoscaling Inference

Scale on the right signal.



GPU cold starts take minutes — keep a warm minimum and scale on queue depth, not CPU.

STEP 6/6

Latency vs Throughput

You can usually have one.

- Latency: time per request. Throughput: requests per second. Batching trades the first for the second.
- Quantization (FP16/INT8) cuts both latency and memory — nearly free speedup for inference.
- Speculative decoding: a small model drafts, the big model verifies — faster without quality loss.
- Set SLOs first (p99 < 200ms?), then choose batching, precision, and replicas to meet them.

STEP ✓/✓

Key Takeaways

Model Serving — Remember

- Batch when you can (cheap); serve real-time only when latency SLOs demand it.
- Use a real inference server (Triton, vLLM) — not a bare web framework.
- Continuous batching + PagedAttention unlock high-throughput LLM serving.
- KV cache makes decode memory-bound; long contexts need cache management.
- Autoscale on queue depth/latency with warm minimums; quantize for free speed.

Keep exploring

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev